

Probabilistic Day-Ahead Electricity Price Forecasting for the Swedish Bidding Zones SE1–SE4

Process, decisions and results: milestones M0–M5

Pipeline `pricefc` · `saldestechnology/ml-electro-mlops`

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Abstract

We built a fully versioned pipeline that forecasts hourly day-ahead electricity prices for the four Swedish bidding zones, issued at 09:00 Europe/Stockholm on day D for every hour of $D+1$, as seven quantiles. All data, datasets, models and evaluations are tracked in MLflow with lineage back to immutable raw snapshots and the git commit. Every input carries an explicit availability rule and every dataset build passes an automated leakage audit. Over a one-year out-of-sample backtest (3 Oct 2025 – 2 Oct 2026, true-lead weather forecasts), a tuned and recalibrated LightGBM quantile model beats the strongest classical baseline (MSTL+ETS) in every zone by 0.78–1.43 EUR/MWh mean pinball loss (13–23%), with block-bootstrap confidence intervals that exclude zero. The pretrained foundation model TimesFM 3.0 with weather and calendar covariates is better still (a further 12–20% over LightGBM, significant in every zone), but its weights are licensed for non-commercial use only, so it serves as a benchmark rather than a deployable model. The deployable TimesFM 2.5 is weaker than LightGBM overall, yet makes complementary errors (it halves the night-hour error), which motivates the ensemble milestone (M8).

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1 Problem and scope

Forecasting task. Nordic day-ahead prices for delivery day $D+1$ are set in the SDAC auction at noon CET on day D and published around 12:45–13:00. A forecast that is useful for bidding must therefore be issued *before* the auction: our origin is 09:00 Europe/Stockholm on day D , and the target is every hour of local day $D+1$ (23, 24 or 25 hours, depending on daylight-saving changes). Forecasts are probabilistic: quantiles $\tau \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$.

Zones and resolution. Sweden has four bidding zones with separate prices: SE1 (north) to SE4 (south, most coupled to the continent and the most volatile). On 1 October 2025 the market moved from hourly to 15-minute market time units; we aggregate to hourly means so the series stays homogeneous, and record the source resolution per row.

Constraints. Work proceeds in milestones (M0 skeleton, M1 ingestion, M2 datasets, M3 backtesting and baselines, M4 LightGBM, M5 TimesFM, M6 registry and serving, M7 cloud GPU, M8 ensembles). All computation so far ran on a laptop (Apple M1 Pro, 32 GB). A \$25 RunPod GPU budget is reserved for foundation-model fine-tuning and may only be spent with the owner’s explicit approval; none has been spent.

2 Architecture and engineering principles

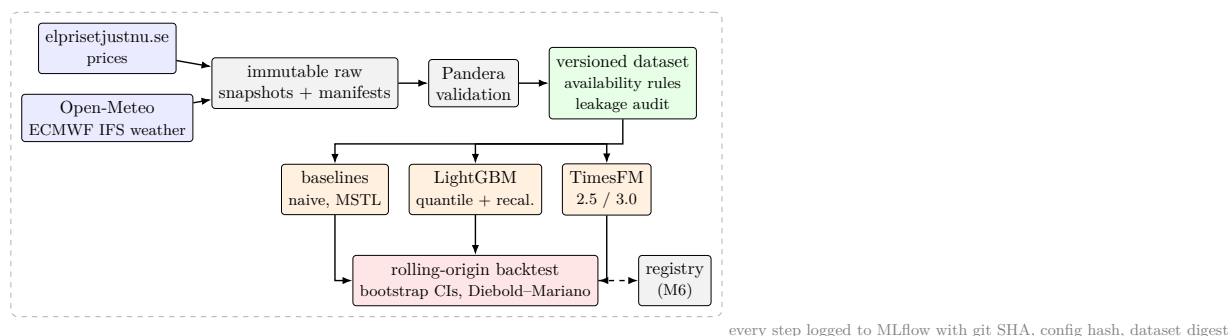


Figure 1: Pipeline overview. Solid arrows exist today; the registry is milestone M6.

- **Reproducibility.** Python 3.12 managed by `uv` with a committed lockfile; YAML configs validated by strict pydantic models and hashed; MLflow (SQLite backend) with required tags on every run: zone, resolution, weather source, dataset version, git SHA and dirty flag, model family, compute environment, config hash and pipeline stage.

- **Immutability.** Raw pulls are written read-only, one directory per source, dataset, key and pull time, with a manifest (request, row counts, hashes, validation result, git SHA). Snapshots are never overwritten; invalid ones are kept and marked, so the record of what an API returned is preserved.
- **Quality gates.** GitHub Actions runs `ruff`, `mypy -strict` and `pytest` (116 offline tests, including Hypothesis property tests) on every push; live API tests are opt-in.
- **Scale.** 29 commits, about 5,100 lines of library code and 1,900 lines of tests.

3 Data (M1)

3.1 Prices

The specification named ENTSO-E as the source of record. Its API token requires a manual application, and the owner decided not to apply. We therefore made the price source pluggable (`PriceSource`) and adopted `elprisetjustnu.se`, which publishes one JSON file per zone and local day from 1 November 2021, in EUR/kWh without VAT.

Practical issues found and handled:

- The default Python user agent receives HTTP 403; the client sends its own.
- `time_end` is computed in wall-clock time, so the last slot before the autumn DST change claims 75 minutes. Resolution is derived from the spacing of `time_start` instead, and anomalies are recorded in the manifest.
- Snapshots are per zone and month; ingest resumes by skipping complete, valid months.

Verification against independent sources. Before trusting the series we compared it with Nord Pool (40 zone-days, 3,840 quarter-hours: identical to the cent) and with Energy-Charts (12 stress days $\times$ 4 zones including DST days in both resolution eras, the market-time-unit switch and the 2022 crisis peak: 47 of 48 zone-days identical). Energy-Charts’ licence permits private use only, so it was used for this check and never as a data source.

The one mismatch exposed a **source defect**: on five days (SE3 2021-11-04; SE2 2022-09-04, 09-10, 09-15, 10-20) the source dropped one hour and shifted every later hour one slot earlier, with errors up to 139 EUR/MWh. Validation caught all five (23 rows on a 24-hour day). These values are wrong rather than incomplete, so they are *never repaired*: a `known_bad_days` list drops them at ingest while still recording how many rows the source returned, so a later correction would be visible.

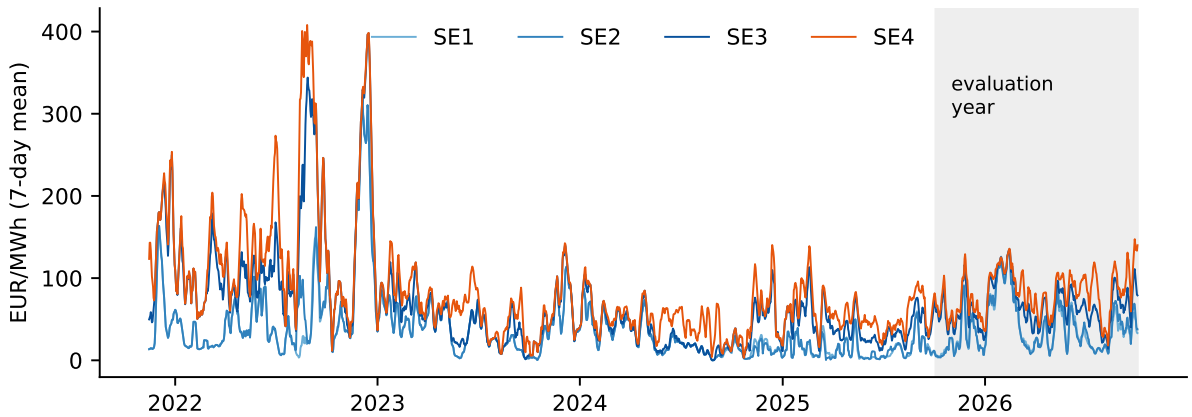


Figure 2: Daily mean day-ahead price (7-day rolling mean), all zones. The 2022 energy crisis dominates the early history; the shaded year is the out-of-sample evaluation window.

3.2 Weather

Weather comes from Open-Meteo, pinned to the **ECMWF IFS** model. The default `best_match` silently switches between underlying models and lacks 100 m wind in the archive, which would break reproducibility. Two endpoints are used:

- *Historical Forecast* (2021 onwards): a “stitched” series of short-range forecasts, used for training. Its exact issue time is unknown.
- *Previous Runs* (from 1–2 October 2025): forecasts as they were issued N days ahead. This is what a live system would have had, so it is used for evaluation.

Each zone uses 3–5 locations (SE4 includes Danish and German points), eight variables, plus zone-level aggregates. The Previous Runs archive has gaps between late March and early May 2026, which validation tolerates per endpoint (`max_null_frac`) and records per column; nothing is imputed silently.

4 Datasets and leakage control (M2)

4.1 Availability rules

Every input source declares when each value becomes known, and feature builders only see an *as-of* view of the data at the origin:

- **Prices** for delivery day X are known from $X-1$ 13:00 local time (conservative relative to the 12:45 publication). At the origin, prices up to the end of day D are therefore known; those of $D+1$ are the target.
- **Weather** valid at time T is known from $T - 48\text{ h} + 8\text{ h}$ (day-2 lead plus publication delay). Open-Meteo defines the day-1 lead as issued 24 h before the valid time; with the delay it would arrive *after* the origin for most hours of $D+1$, so the stricter day-2 lead is used.

4.2 Leakage audit

Every dataset build runs an audit: for seeded random origins, both ends of the range and every origin whose target day has 23 or 25 hours, all data that is unavailable at the origin is perturbed and every feature must stay bit-identical. The perturbation must also change the target, so the test cannot pass vacuously. A build that fails the audit is rejected. A Hypothesis property test and deliberately injected leaks (which the audit must catch) guard the audit itself. Later we added a second guard: the backtest harness strips the target and its metadata before calling a model’s `predict`, enforced by a test.

4.3 Features and versioning

Rows are (origin day, target hour). Features: calendar (hour, weekday, Swedish public, de facto and half holidays, bridge days, DST), own-price lags and rolling statistics up to the end of day D , neighbouring-zone prices, and weather at each location plus zone aggregates (108 features for SE3). Datasets are stored as `data/datasets/{zone}/hourly/{date}-{digest}/` with a lineage manifest listing every contributing raw snapshot, and logged to MLflow as dataset inputs.

A locale bug caught by CI. The `holidays` package localises holiday names from the system locale, and the calendar code matched on English names. Under a C or Swedish locale every Sunday became a public holiday. The language is now pinned and a test builds the calendar under three locales.

4.4 The train/evaluation weather gap

Training uses the stitched weather series, evaluation the true day-2 forecasts. On the overlapping SE3 year they agree closely for temperature ($r = 0.996$) and radiation ($r = 0.991$) but less for 100 m wind ($r = 0.939$, MAE 0.6 m/s), the most price-relevant variable (Figure 3). Models trained on the stitched series should therefore be expected to be over-confident in evaluation, which is exactly what we later observed (Section 6.2).

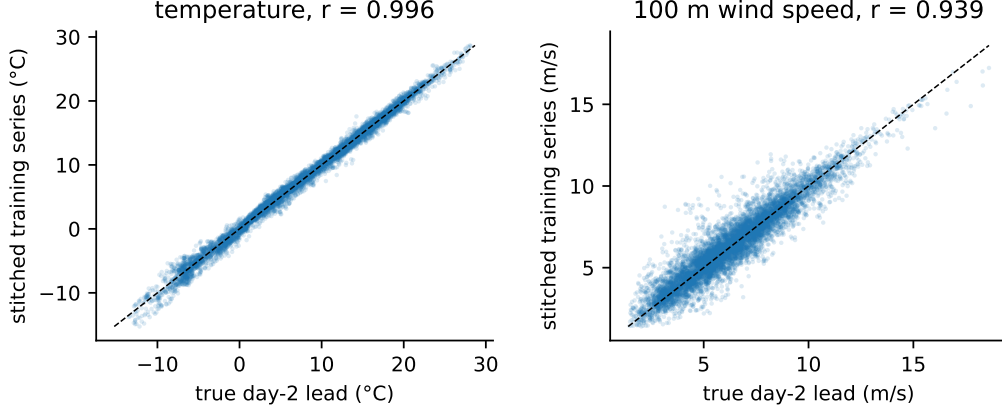


Figure 3: Stitched training weather versus the true day-2 forecast (SE3 zone means, one year).

5 Evaluation methodology (M3)

Rolling-origin backtest. For each of the 365 origins from 3 October 2025 to 2 October 2026, models are trained on stitched rows whose target day is at most D (i.e. published before the origin) and predict the true-lead rows of $D+1$. Models refit monthly by default; models that need the most recent history refit at every origin. Quantile crossings are repaired by sorting.

Metrics. The primary metric is the pinball loss averaged over the seven quantiles,

$$L_\tau(y, q) = \max(\tau(y - q), (\tau - 1)(y - q)), \quad \bar{L} = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \text{mean}_t L_\tau(y_t, q_{\tau,t}), \quad (1)$$

complemented by MAE of the median, bias, sMAPE (denominator floored at 1 EUR/MWh because of negative prices), empirical coverage of the 80% and 90% intervals, and slices by hour, weekday, season, spike days and negative-price hours.

Statistical comparison. Losses are aggregated per day. Confidence intervals come from a circular block bootstrap (7-day blocks, 2,000 resamples) of the daily mean loss and of paired daily differences between two models. The Diebold–Mariano test with the Harvey–Leybourne–Newbold small-sample correction is applied to the daily loss differential d_t :

$$\text{DM} = \frac{\bar{d}}{\sqrt{\hat{\gamma}_0/n}}, \quad \text{DM}_{\text{HLN}} = \text{DM} \sqrt{\frac{n+1-2h+h(h-1)/n}{n}}, \quad h = 1, \quad (2)$$

compared with a t_{n-1} distribution.

Lesson: biased development samples. A quick development mode evaluates every k -th origin. With $k = 7$ every target was a Saturday, which reversed the ranking of the 1-day and 7-day naive models. The default is now $k = 5$, and multiples of 7 are rejected by config validation.

A second lesson came later (M5): a 73-origin development sample ranked TimesFM 2.5 level with LightGBM, whereas the full year showed it clearly worse in three zones. Decisions are only taken on full-year runs.

Baselines. Seasonal naive (same hour 7 days and 1 day earlier), the EPF “weekday” naive, and StatsForecast MSTL with daily and weekly seasonality and an AutoETS trend (56-day context). Naive quantiles are empirical per-hour quantiles of each method’s own recent errors. Table 1 shows that MSTL is much stronger than the naive methods; it became the bar for M4.

Table 1: Baselines, 365 origins: mean pinball loss (EUR/MWh) with 95% block-bootstrap CI for MSTL, and MSTL skill relative to the 7-day naive reference.

Zone	MSTL+ETS	Naive 1d	Weekday naive	Naive 7d	MSTL skill
SE1	5.95 (5.31–6.65)	7.70	8.41	11.12	0.47
SE2	6.22 (5.53–6.97)	8.02	8.80	11.48	0.46
SE3	6.53 (6.09–6.98)	9.15	9.16	11.61	0.44
SE4	8.35 (7.63–9.14)	11.41	11.52	13.69	0.39

6 LightGBM quantile model (M4)

6.1 Model and tuning

One LightGBM regressor per quantile with the quantile objective, on all dataset features. Hyperparameters were tuned per zone with Optuna (seeded TPE sampler, median pruner, 40 trials) on four expanding-window folds of 91 days that all end *before* the evaluation year; the code asserts that tuning never touches the evaluation window. The objective is mean pinball loss over $\tau \in \{0.1, 0.5, 0.9\}$, and the number of trees is the median early-stopping iteration of the best trial. Interestingly, SE3 prefers large trees (252 leaves), whereas SE1, SE2 and SE4 prefer small, heavily regularised trees (17–29 leaves). Tuning improved SE3 only modestly over sensible defaults (−0.09 EUR/MWh, CI [−0.19, +0.005]).

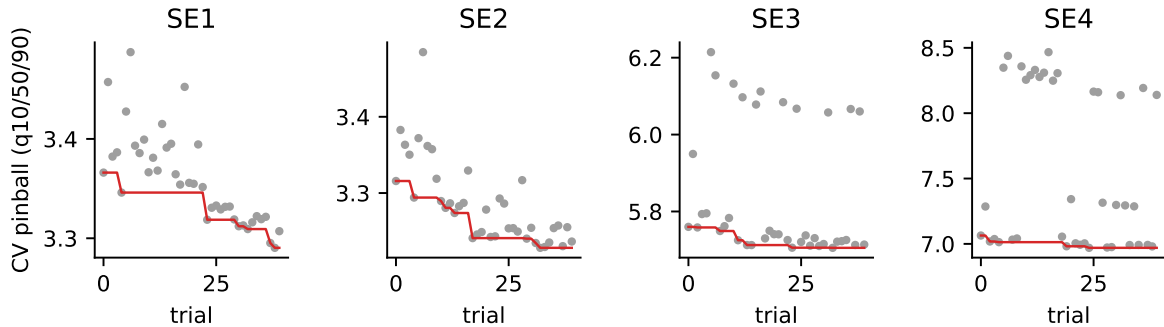


Figure 4: Optuna studies: cross-validated pinball loss per trial (grey) and best so far (red).

6.2 Over-confidence and recalibration

On true-lead data the raw model was over-confident: its 80% and 90% intervals covered only 72% and 84% of SE3 prices, and the median ran 5.7 EUR/MWh low. This matches the weather gap of Figure 3: the model learned from more accurate weather than it receives live. We added a model-agnostic wrapper that, at every origin, shifts each quantile by the τ -quantile of the model’s

own residuals over the last 28 published days:

$$\hat{q}_\tau^{\text{cal}} = \hat{q}_\tau + Q_\tau(\{y_s - \hat{q}_{\tau,s} : s \text{ in the last 28 published days}\}). \quad (3)$$

Only realised prices that are published before the origin enter the offsets. Coverage moved to 77.6%/87.9% and the pinball loss improved slightly (Figure 5); a 14-day window was too noisy. The first week of a deployment is uncalibrated (cold start).

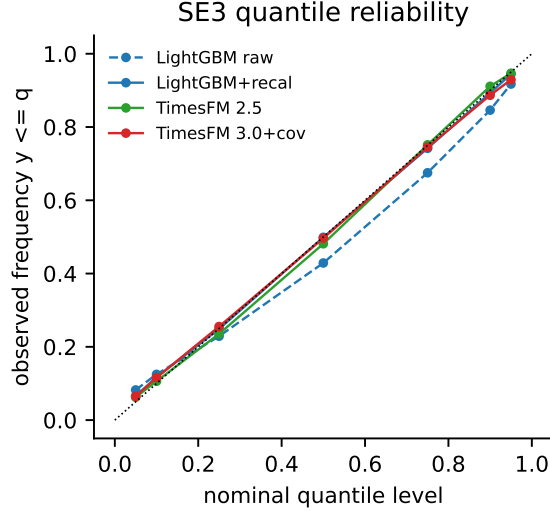


Figure 5: Reliability of the SE3 quantiles: the fraction of prices below each forecast quantile versus its nominal level. The dashed line is raw LightGBM.

6.3 Results

Recalibrated LightGBM beats MSTL in every zone (Table 2). Further checks on SE3:

- **Seed variance:** six seeds give 5.730 ± 0.009 , about 100 times smaller than the margin over MSTL.
- **Crisis years:** training only from July 2023 is clearly worse (6.09 vs 5.75), so the 2021–2023 crisis history is kept.
- **Importance:** own price history 55% of gain (the price 24 hours earlier alone 26%), weather 27% with 100 m wind leading, calendar 12%, neighbouring zones 6% (Figure 6).
- **Remaining weakness:** the median still under-forecasts by 2–4 EUR/MWh on average; the residual-quantile offset does not target mean bias and price spikes pull the mean above the median.

Table 2: LightGBM, 365 origins per zone: mean pinball loss (EUR/MWh), paired difference to MSTL with 95% block-bootstrap CI, Diebold–Mariano p-value, interval coverage and median bias.

Zone	LGBM+recal	LGBM raw	MSTL	Naive 7d	Δ vs MSTL [95% CI]	DM p	Cov. 80/90	Bias
SE1	5.053	5.060	5.950	11.117	−0.90 [−1.26, −0.54]	$2 \cdot 10^{-7}$	76.9/88.3	−3.4
SE2	4.788	4.790	6.216	11.479	−1.43 [−1.75, −1.11]	$7 \cdot 10^{-15}$	78.4/88.8	−4.0
SE3	5.727	5.753	6.530	11.613	−0.78 [−1.15, −0.40]	$< 10^{-3}$	77.6/87.9	−2.1
SE4	6.920	7.011	8.351	13.691	−1.43 [−1.89, −1.00]	$8 \cdot 10^{-12}$	77.7/88.7	−1.7

By the promotion criteria of the specification (better mean pinball than the incumbent, paired CI excluding zero or significant DM test, coverage within tolerance, clean leakage audit), the

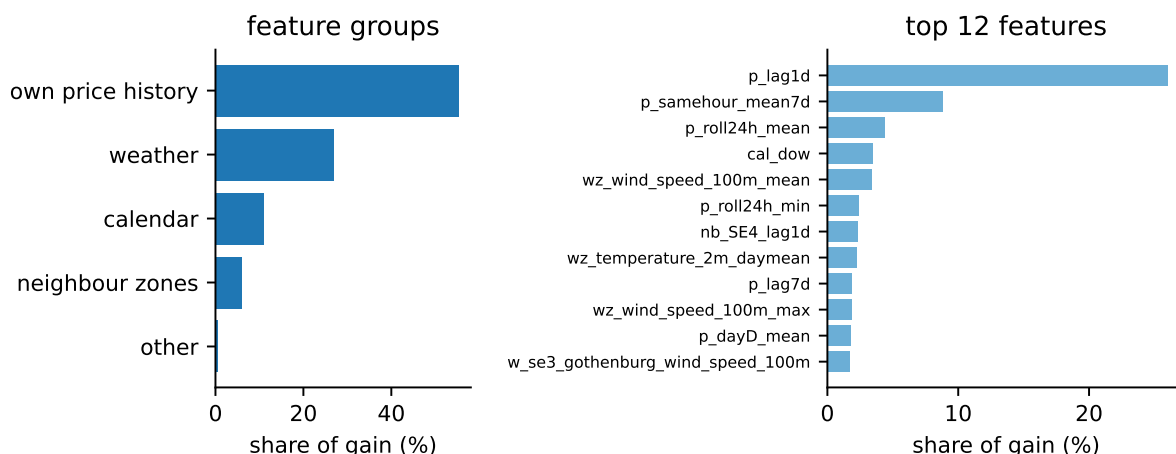


Figure 6: SE3 LightGBM feature importance (gain of the median model, averaged over monthly refits).

recalibrated LightGBM is the champion candidate in every zone. Formal registration requires the model wrapper of M6.

7 Foundation model TimesFM (M5)

7.1 Versions and licensing

Checking the current release before building (as the specification requires) found two relevant checkpoints, both pinned by Hugging Face revision hash:

- **TimesFM 2.5** (200M parameters, Apache-2.0): context up to 16k points, decile outputs, covariates through “XReg” (an in-context linear regression, with TimesFM forecasting its residuals).
- **TimesFM 3.0** (released August 2026): native past and future covariates and an MLX backend for Apple silicon. Its weights are licensed for *non-commercial, non-production use only*, and fine-tuned versions are derivatives under the same terms.

The owner decided to evaluate 2.5 as the deployable candidate and 3.0 as a research benchmark. The adapter refuses to load 3.0 unless `accept_noncommercial_licence` is set, and its runs are tagged `deployable=false` so they can never be promoted.

7.2 Adapter

TimesFM plugs into the same harness as every other model. At each origin the context is the last 2,048 published hours (ending at local midnight before $D+1$), and the forecast covers exactly the target hours. Covariates are the zone-mean temperature, 100 m wind, radiation and cloud cover plus calendar features; their future values come from the leakage-audited feature rows. TimesFM emits deciles, which are linearly interpolated to our quantile levels, with the 5% and 95% tails extrapolated from the median using normal-distribution scaling.

Smoke tests on the laptop: 2.5 takes 0.24 s per forecast on CPU (the Apple GPU was slower); 3.0 on MLX takes 0.05 s. A full-year backtest takes 1.5–5 minutes per model, so no GPU was needed. Two problems surfaced: TimesFM 2.5 clamps forecasts at zero whenever the context has no negative prices unless `infer_is_positive` is disabled, and torch inference *deadlocked* after a LightGBM fit in the same process, because the two libraries bundle separate copies of the OpenMP runtime on macOS. Restricting torch to one thread removed the deadlock at little cost.

7.3 Results

Table 3: TimesFM versus recalibrated LightGBM, 365 origins per zone: mean pinball loss (EUR/MWh) and paired difference to LightGBM with 95% block-bootstrap CI.

Zone	LGBM+recal	TimesFM 3.0 + covariates		TimesFM 3.0	TimesFM 2.5 (deployable)	
		pinball	Δ [95% CI]		pinball	Δ [95% CI]
SE1	5.053	4.054	-1.00 [-1.28, -0.72]	4.792	4.927	-0.13 [-0.42, +0.19]
SE2	4.788	3.866	-0.92 [-1.21, -0.66]	5.082	5.188	+0.40 [+0.14, +0.67]
SE3	5.727	4.989	-0.74 [-1.01, -0.45]	5.757	6.118	+0.39 [+0.07, +0.71]
SE4	6.920	6.059	-0.86 [-1.11, -0.62]	7.110	7.505	+0.59 [+0.24, +0.95]

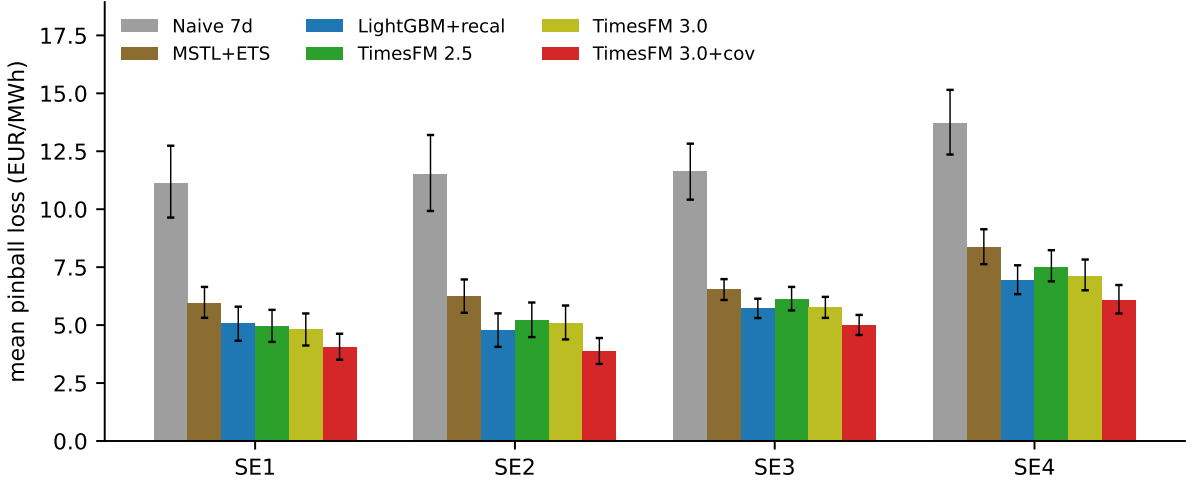


Figure 7: Mean pinball loss per zone and model over the evaluation year, with 95% block-bootstrap confidence intervals.

- **TimesFM 3.0 with covariates is the best model in every zone**, 12–20% better than LightGBM (DM $p < 0.001$ everywhere), and its intervals are close to nominal without any recalibration (coverage 76–80% / 86–88%). Without covariates, 3.0 is roughly level with LightGBM: the covariates matter.
- **TimesFM 2.5 alone is worse than LightGBM** in SE2–SE4 and level in SE1. Adding XReg covariates made 2.5 *worse*; a linear in-context regression does not capture the weather–price relation. Recalibration does not help either TimesFM model.
- **The errors are complementary** (Figure 8). TimesFM 2.5, and even MSTL, beat LightGBM clearly in the first hours of the delivery day: models that read the series up to its last published hour (23:00 on day D) carry the overnight level forward, whereas LightGBM’s nearest same-hour lag is 24 hours old. LightGBM wins from the morning ramp onwards and on spike days, where weather drives prices.

7.4 Fine-tuning decision

Fine-tuning on a rented GPU is deferred. TimesFM 3.0 already wins but cannot be deployed, and a fine-tune would inherit its licence. TimesFM 2.5 meets the specification’s precondition (better on some slices), but combining it with LightGBM costs nothing and should be tried first. Fine-tuning 2.5 will only be proposed, with a cost estimate for the owner’s approval, if a clear gap to TimesFM 3.0 remains after the ensemble.

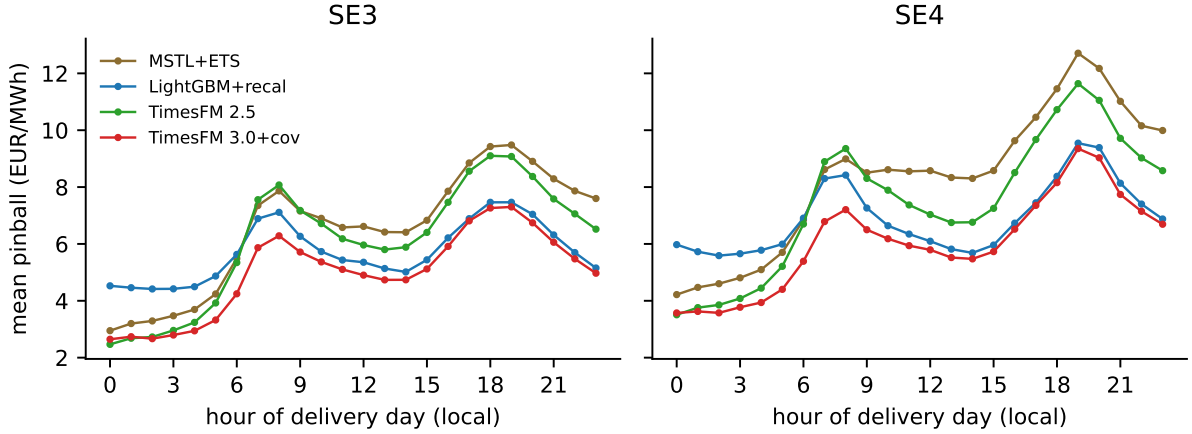


Figure 8: Mean pinball loss by hour of the delivery day.

8 Issues found and fixed along the way

Issue	Resolution
Source shifted hours on 5 days	Verified against Energy-Charts; excluded via <code>known_bad_days</code> , never repaired.
Open-Meteo <code>best_match</code> switches models	Pinned ECMWF IFS for training and evaluation.
Day-1 weather lead arrives after the origin	Day-2 lead with explicit availability rule.
Holiday names depend on system locale	Language pinned; test under three locales.
Development sample hit only Saturdays	Stride 5 by default; multiples of 7 rejected.
Model variants overwrote each other in comparisons	Keyed by full variant spec; regression test.
<code>predict</code> received the target column	Harness strips target and metadata; test.
Raw LightGBM over-confident on live weather	Rolling 28-day quantile recalibration.
TimesFM clamped forecasts at zero	<code>infer_is_positive</code> disabled.
LightGBM + torch	Torch limited to one thread.
OpenMP deadlock (macOS)	

9 Conclusions and next steps

1. A leakage-audited, fully versioned pipeline produces calibrated quantile forecasts for all four zones, with every number in this report traceable to an MLflow run.
2. The deployable champion candidate is the tuned, recalibrated LightGBM: 13–23% better than MSTL and 49–58% better than the seasonal naive reference, significant in every zone.
3. TimesFM 3.0 with covariates shows a further 12–20% of headroom, but only as a non-deployable benchmark.
4. TimesFM 2.5 and LightGBM make complementary errors; the next step (M8) is an ensemble of the two deployable models, weighted per hour of day and fitted on past backtest forecasts

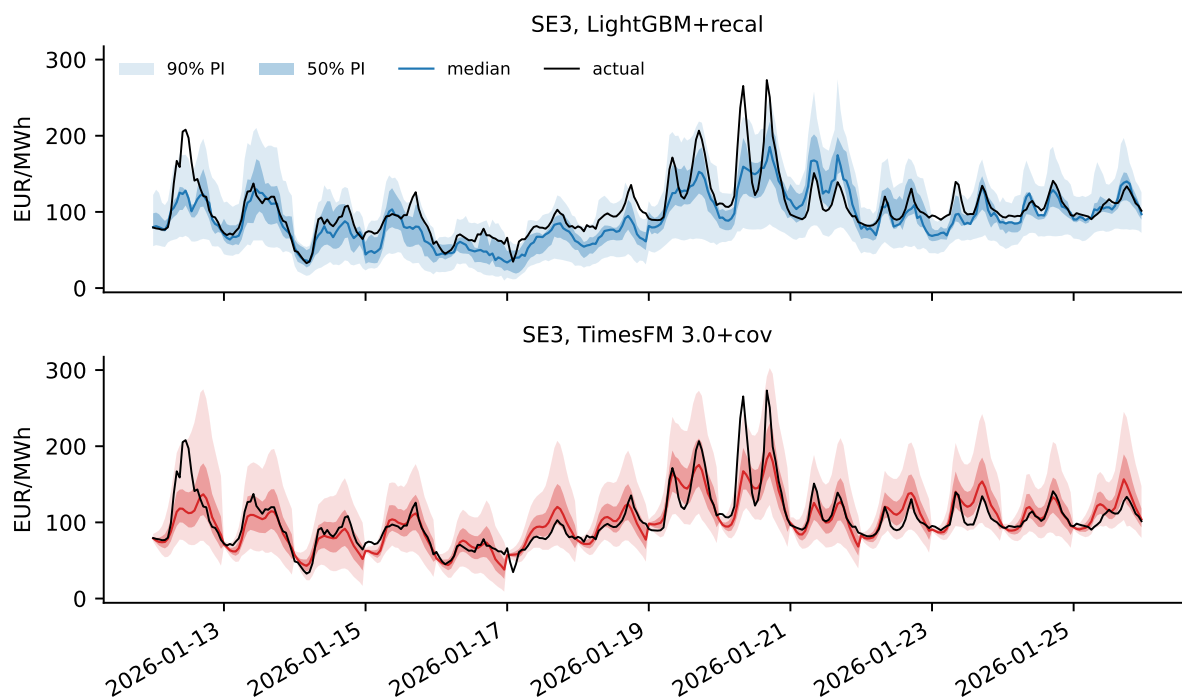


Figure 9: Two winter weeks in SE3: 50% and 90% prediction intervals, median and actual price for recalibrated LightGBM (top) and TimesFM 3.0 with covariates (bottom).

only, to see how much of the gap to TimesFM 3.0 can be closed without licence restrictions.

5. A cheap LightGBM improvement follows from Figure 8: features describing the last published hours of day D should fix its night-hour weakness.
6. M6 will wrap the chosen model for the MLflow registry and serve the daily 09:00 forecast.

A Reproducing the results

```
uv sync --extra timesfm                                # environment from the lockfile
uv run python -m pricefc ingest prices                  # elprisetjustnu.se, resumable
uv run python -m pricefc ingest weather                 # Open-Meteo ECMWF IFS
uv run python -m pricefc dataset build stitched -z SE3
uv run python -m pricefc dataset build true_lead -z SE3
uv run python -m pricefc tune -z SE3                    # Optuna, before eval year
uv run python -m pricefc backtest run -z SE3 \
  -m mstl_ets -m "lightgbm_tuned_SE3@calibration=28" \
  -m timesfm25 -m timesfm3_cov
uv run python -m pricefc backtest compare <run ids> --reference mstl_ets
uv run python docs/report/make_figures.py              # figures in this report
```

Dataset versions used: stitched 20261004-* and true-lead 20261004-* per zone (SE3: 959ffb and 220df7). The decision log docs/decisions.md records every choice with its date and evidence.